

Module 3

Sequential Logic Design

Motivation:

Sequential elements, latches and flip flops dissipate power when there is switching in their internal capacitance. This may happen with every clock transition/pulse into the sequential element. Sometimes the sequential elements need to change their state, but sometimes they retain their state and their output remains the same, before and after the clock pulse. This leads to unnecessary dissipation of power due to clock transition. If flip flops are designed in such a way that they are able to gate the clock with respect to their own internal data path, only to use the clock when needed, power dissipation can be brought down.

Syllabus:

Lecture no	Content	Duration (Hr)	Self-Study (Hrs)
1	Concepts of Flip Flops: SR, D, J-K, T, Truth Tables and Excitation Tables of all types	2	2
2	Race around condition, Master Slave J-K Flip Flops	2	2
3	Timing Diagram, Flip-flop conversion	2	2
4	Introduction to Counters: Design of Asynchronous and Synchronous Counters, Modulus of the Counters	2	2
5	UP- DOWN counter, Ring and twisted ring/Johnson Counter, sequence generator	2	2
6	Introduction to Registers: SISO, SIPO, PIPO, PISO	2	2

Learning Objectives:

Learners shall be able to:

1. Design and analyze sequential circuits
2. To explain the concept of flip-flops like SR, D, J-K and T with truth tables and excitation tables.
3. To explain the concept of Race around condition
4. To explain the concept of Master Slave J-K Flip Flops
5. To construct Timing Diagram and to convert one to another Flip-flop

6. To design of Asynchronous and Synchronous Counters and Modulus of the Counters.
7. To explain different counters like UP- DOWN counter, Ring and twisted ring/Johnson Counter, sequence generator.
8. To explain different Registers like SISO, SIPO, PIPO, PISO.

Theoretical Background:

Digital electronics is classified into combinational logic and sequential logic. Combinational logic output depends on the inputs levels, whereas sequential logic output depends on stored levels and also the input levels. The memory elements are devices capable of storing binary info. The binary information stored in the memory elements at any given time defines the state of the sequential circuit. The input and the present state of the memory element determine the output. Memory elements next state is also a function of external inputs and present state. A sequential circuit is specified by a time sequence of inputs, outputs, and internal states. Examples of sequential circuits are Flip-Flops, latches, counters, registers, and time state generators.

"Flip-flop" is the common name given to two-state devices which offer basic memory for sequential logic operations. Flip-flops are heavily used for digital data storage and transfer and are commonly used in banks called "registers" for the storage of binary numerical data.

Latches are flip-flops for which the timing of the output changes is not controlled. For a latch, the output essentially responds immediately to changes on the input lines (and possibly the presence of a clock pulse). A flip-flop is designed to change its output at the edge of a controlling clock signal.

Key Definitions:

Flip Flop: Flip-flop is the common name given to two-state devices which offer basic memory for sequential logic operations. Flip-flops are heavily used for digital data storage and transfer and are commonly used in banks called "registers" for the storage of binary numerical data.

Excitation Table: An excitation table shows the minimum inputs that are necessary to generate a particular next state (in other words, to "excite" it to the next state) when the current state is known. They are similar to truth tables and state tables, but rearrange the data so that the current state and next state are next to each other on the left-hand side of the table, and the inputs needed to make that state change happen are shown on the right side of the table.

Race around Condition: For J-K flip-flop, if $J=K=1$, and if $clk=1$ for a long period of time, then Q output will toggle as long as CLK is high, which makes the output of the flip-flop unstable or uncertain. This problem is called race around condition in J-K flip-flop.

SR Flip Flop: SR flip-flop is basically a one-bit memory bistable device that has two inputs, one which will "SET" the device (meaning the output = "1"), and is labelled S and one which will "RESET" the device (meaning the output = "0"), labelled R.

T Flip Flop: T flip – flop is an edge triggered device i.e. the low to high or high to low transitions on a clock signal of narrow triggers that is provided as input will cause the change in output state of flip – flop.

D Flip Flop: The D-type flip-flop is a modified Set-Reset flip-flop with the addition of an inverter to prevent the S and R inputs from being at the same logic level

JK Flip Flop: The JK flip flop is basically a gated SR flip-flop with the addition of a clock input circuitry that prevents the illegal or invalid output condition that can occur when both inputs S and R are equal to logic level "1".

Master-Slave JK Flip Flop: The master-slave flip-flop eliminates all the timing problems by using two SR flip-flops connected together in a series configuration. One flip-flop acts as the "Master" circuit, which triggers on the leading edge of the clock pulse while the other acts as the "Slave" circuit, which triggers on the falling edge of the clock pulse. This results in the

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two sections, the master section and the slave section being enabled during opposite half-cycles of the clock signal.

Truth Tables: The table used to represent the boolean expression of a logic gate function is commonly called a Truth Table.

Excitation: an 'EXCITATION TABLE' shows the minimum inputs that are necessary to generate a particular next state (in other words, to "excite" it to the next state) when the current state is known. For a given combination of present state Q_n and next state Q_{n+1} , excitation table tell the inputs required.

Timing Diagram: Timing diagrams are UML interaction diagrams used to show interactions when a primary purpose of the diagram is to reason about time. Timing diagrams focus on conditions changing within and among lifelines along a linear time axis. In a timing diagram, time passes on the x-axis from left to right, with different components of the system that interact with each other on the y-axis.

Counter:

A counter is a device which can count any particular event on the basis of how many times the particular event(s) is occurred. In a digital logic system or computers, this counter can count and store the number of time any particular event or process have occurred, depending on a clock signal.

Asynchronous Counters:

Asynchronous counters are those whose output is free from the clock signal. Because the flip flops in asynchronous counters are supplied with different clock signals, there may be delay in producing output.

Synchronous Counters:

A synchronous counter, in contrast to an asynchronous counter, is one whose output bits change state simultaneously, with no ripple.

Modulus of the Counters:

The Modulus (or MOD-number) of a counter is the total number of unique states it passes through in one complete counting cycle with a mod-n counter being described also as a divide-by-n counter. The modulus of a counter is given as: 2^n where n = number of flip-flops.

UP- DOWN counter:

An up-down counter is a combination of an up-counter and a down-counter. It can count in both directions, increasing as well as decreasing. Depending on the type of clock inputs, counters are of two types: asynchronous counters and synchronous counters.

Ring and twisted ring/Johnson Counter:

Ring counter is a sequential logic circuit that is constructed using shift register. Same data recirculates in the counter depending on the clock pulse.

Sequence generator:

A sequence generator is one kind of digital logic circuit. The main function of this is to generate a set of outputs. Every output is one of a number of binary or Q-ary logic levels or symbols.

Course Content:

Lecture : 17

Concepts of Flip Flops: SR, D, J-K, T, Truth Tables and Excitation Tables of all types

Flip flops are an application of logic gates. A flip-flop circuit can remain in a binary state indefinitely (as long as power is delivered to the circuit) until directed by an input signal to switch states.

S-R flip-flop stands for SET-RESET flip-flops.

The SET-RESET flip-flop consists of two NOR gates and also two NAND gates.

These flip-flops are also called S-R Latch.

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The design of these flip flops also includes two inputs, called the SET [S] and RESET [R]. There are also two outputs, Q and Q'.

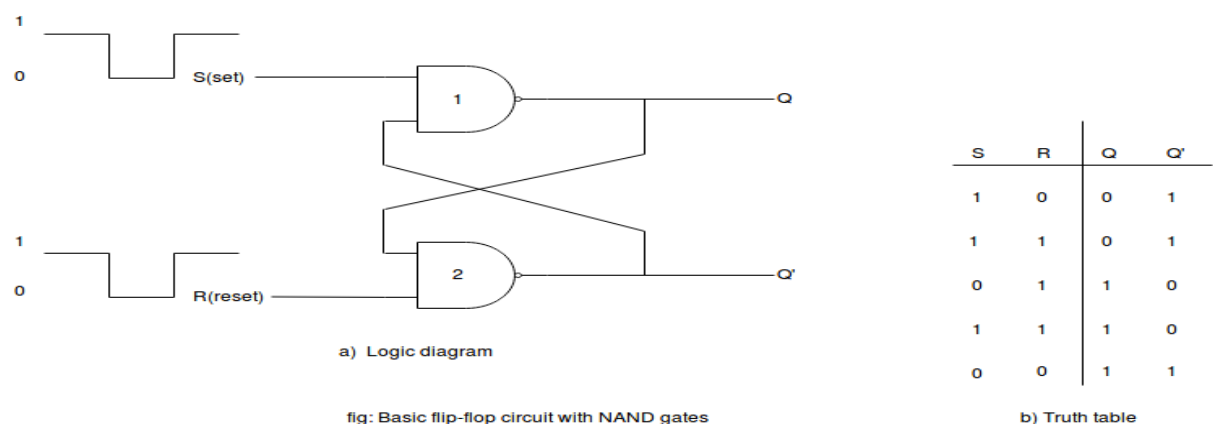
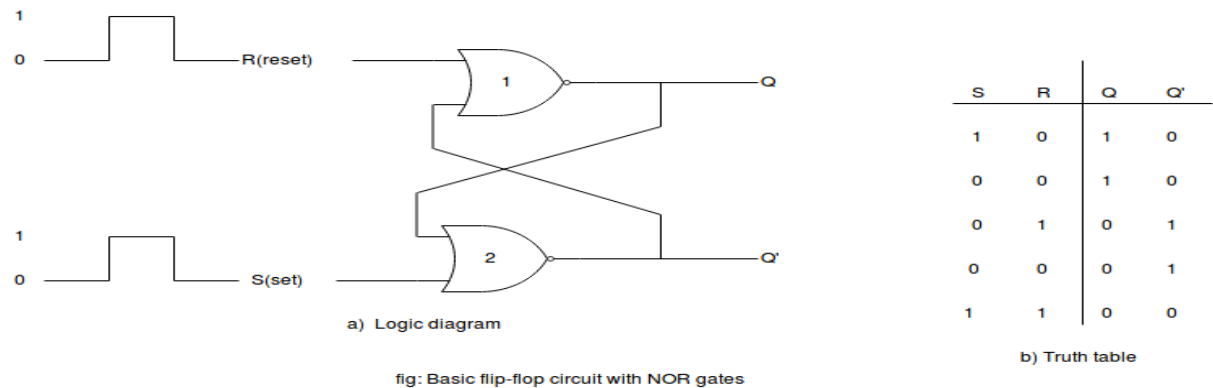


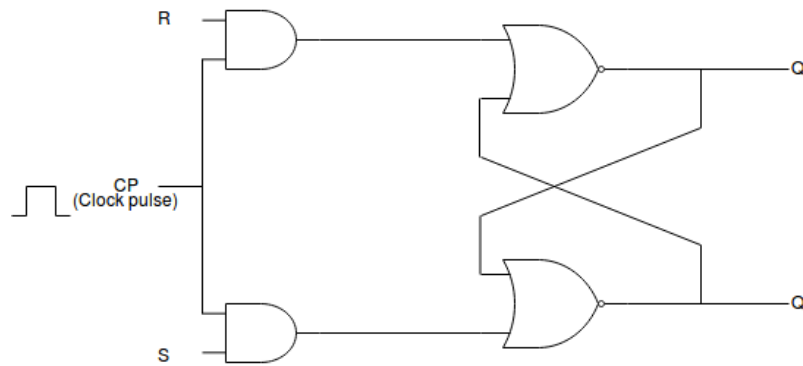
Fig. 3.1.1 basic flip flop with NOR and NAND gates.

Clocked S-R Flip-Flop

The operation of a basic flip-flop can be modified by providing an additional control input that determines when the state of the circuit is to be changed.

The limitation with a S-R flip-flop using NOR and NAND gate is the invalid state. This problem can be overcome by using a stable SR flip-flop that can change outputs when certain invalid states are met, regardless of the condition of either the Set or the Reset inputs.

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a) Logic diagram

fig: Clocked SR flip flop

Q	S	R	Q(t+1)
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	Intermediate
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	Intermediate

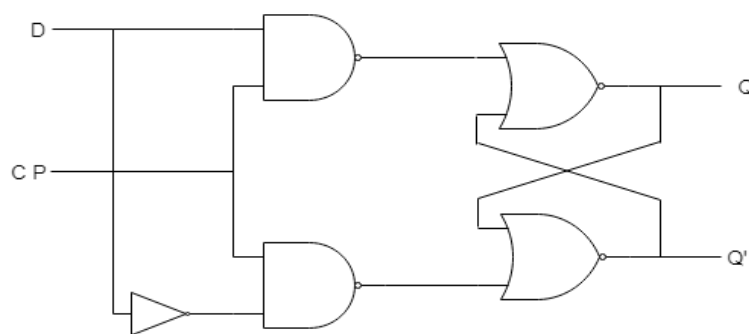
b) Truth table

Fig. 3.1.2 Clocked SR flip flop.

A clock pulse is given to the inputs of the AND Gate. If the value of the clock pulse is '0', the outputs of both the AND Gates remain '0'.

D Flip-Flop

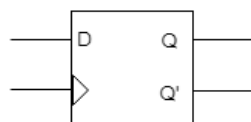
D flip-flop is a slight modification of clocked SR flip-flop.



(a) Logic diagram with Nand gates

Q	D	Q (t + 1)
0	0	0
0	1	1
1	0	0
1	1	1

(c) Transition table



(b) Graphic Symbol

fig. Clocked D flip flop

Fig. 3.1.3 Clocked D flip flop.

From the above figure, you can see that the D input is connected to the S input and the complement of the D input is connected to the R input.

When the value of CP is '1' (HIGH), the flip-flop moves to the SET state if it is '0' (LOW), the flip-flop switches to the CLEAR state.

J-K flip-flop

J-K flip-flop can be considered as a modification of the S-R flip-flop.

The main difference is that the intermediate state is more refined and precise than that of an S-R flip-flop.

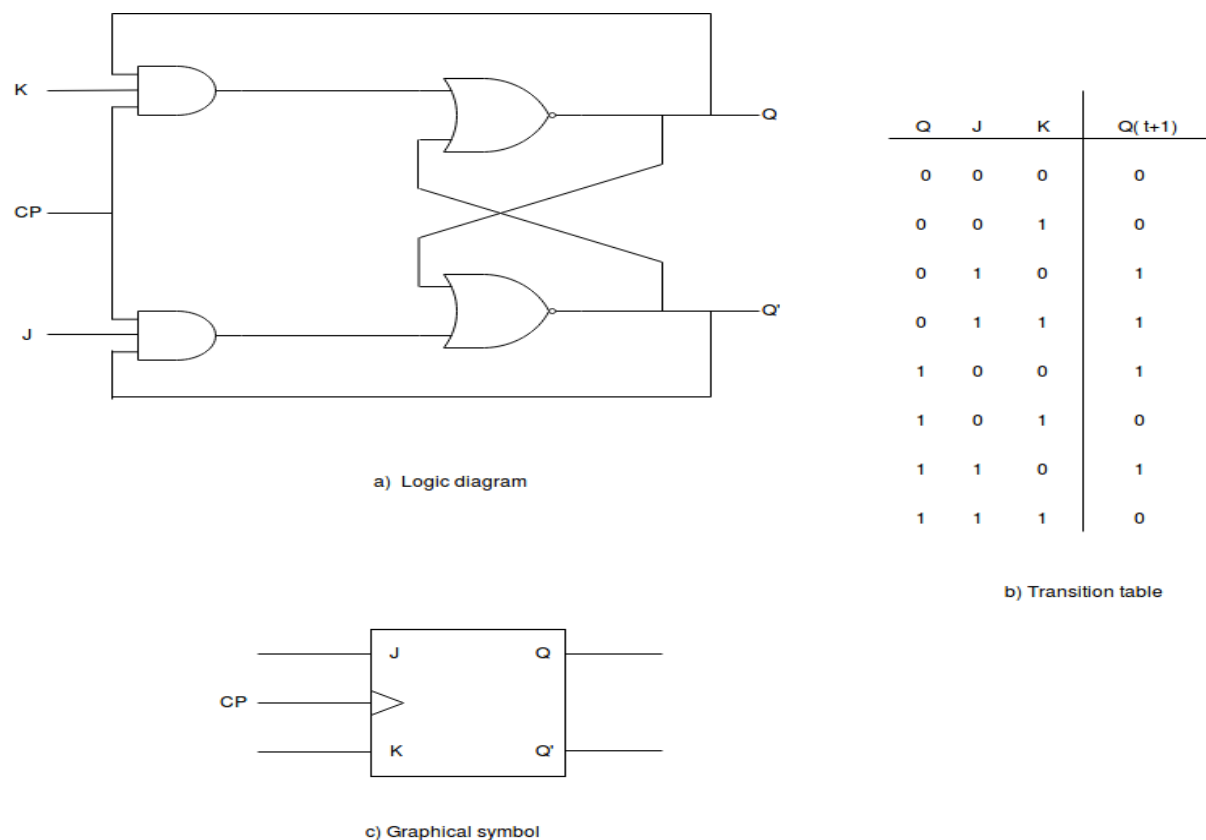


fig. Clocked JK flip flop

Fig. 3.1.4 Clocked JK flip flop.

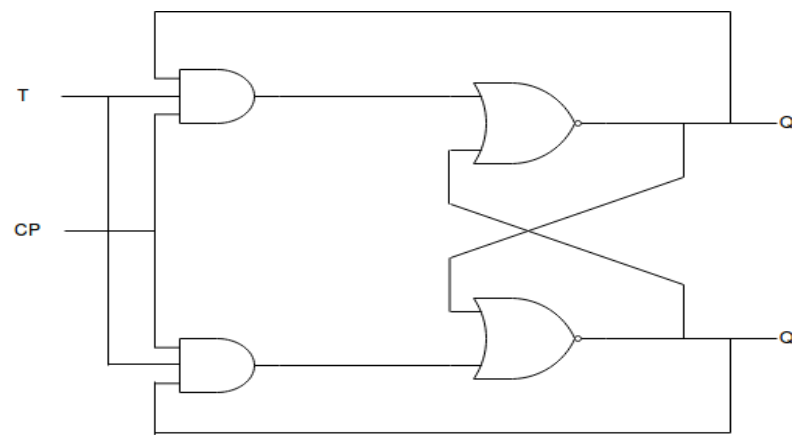
The characteristics of inputs 'J' and 'K' is same as the 'S' and 'R' inputs of the S-R flip-flop.

J stands for SET, and 'K' stands for CLEAR.

When both the inputs J and K have a HIGH state, the flip-flop switches to the complement state, so, for a value of $Q = 1$, it switches to $Q=0$, and for a value of $Q = 0$, it switches to $Q=1$.

T flip-flop

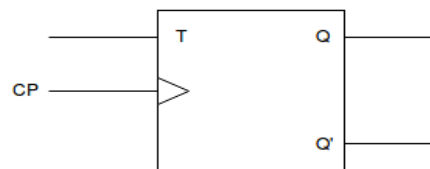
T flip-flop is a much simpler version of the J-K flip-flop.



a) Logic diagram

Q	T	Q(t+1)
0	0	0
0	1	1
1	0	1
1	1	0

b) Transition table



c) Graphical symbol

fig. Clocked T flip flop

Fig. 3.1.5 Clocked T flip flop.

Both the J and K inputs are connected and are also called as a single input J-K Flip-flop.

Triggering of Flip-Flops

The state of the flip-flop is changed by a momentary change in the input signal. This momentary change is known as Trigger, and the transition it causes is said to triggering the flip-flop.

Pulses trigger clocked flip-flops.

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A pulse start from the initial value of '0', goes momentarily to '1', and after a short while, returns to its initial '0' value.

A clock pulse is either positive or negative.

A positive clock source remains at '0' during the interval between pulses and goes to 1 during the occurrence of a pulse.

The pulse goes through two signal transition: from '0' to '1' and return from '1' to '0'.

Definition of clock pulse transition:

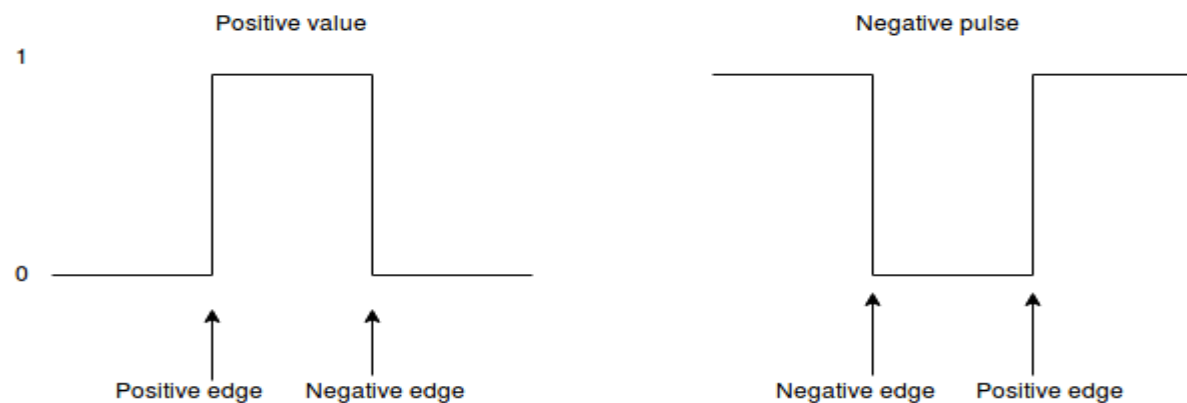


Fig.3.1.6 Clock pulse transition.

The positive transition is defined as a positive edge and the negative transition as a negative edge.

Let's check the take away from this lecture

1) A basic S-R flip-flop can be constructed by cross-coupling of which basic logic gates?

- a) AND or OR gates
- b) XOR or XNOR gates
- c) **NOR or NAND gates**
- d) AND or NOR gates

2) In S-R flip-flop, if $Q = 0$ the output is said to be _____

- a) Set
- b) **Reset**
- c) Previous state
- d) Current state

3) Which of the following is correct for a gated D-type flip-flop?

- a) **The Q output is either SET or RESET as soon as the D input goes HIGH or LOW**

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- b) The output complement follows the input when enabled
 - c) Only one of the inputs can be HIGH at a time
 - d) The output toggles if one of the inputs is held HIGH
- 4) When both inputs of a J-K flip-flop cycle, the output will _____
- a) Be invalid
 - b) Change
 - c) Not change**
 - d) Toggle
- 5) The characteristic of J-K flip-flop is similar to _____
- a) S-R flip-flop**
 - b) D flip-flop
 - c) T flip-flop
 - d) Gated T flip-flop
- 6) How is a J-K flip-flop made to toggle?
- a) $J = 0, K = 0$
 - b) $J = 1, K = 0$
 - c) $J = 0, K = 1$
 - d) $J = 1, K = 1$**

Learning from this lecture: Learners will be able to understand the Concepts of Flip flop ,Truth Tables and Excitation Tables of all types

Lecture : 18

Race around condition, Master Slave J-K Flip Flops

Race around Condition

The following are the two cases where race

1. When the S and R inputs of a SR flip flop are both HIGH and it is known as race condition.

2. For J-K flip-flop, if $J=K=1$, and if clock is high, the output

toggles as long as CLK is high, which makes the output of the flip-flop unstable or uncertain.

This problem is called race around condition in J-K flip-flop. This problem (Race Around

Exercise

- Q.1 Explain SR flip flop with timing diagram.
- Q.2 Explain D flip flop with timing diagram.
- Q.3 Explain J-K flip flop with timing diagram.
- Q.4 Explain T flip flop with timing diagram.

Condition) can be avoided by ensuring that the clock input is at logic “1” only for a very short time. This introduced the concept of Master Slave JK flip flop.

Master Slave J- K Flip Flop

The Master-Slave Flip-Flop is basically a combination of two JK flip-flops connected together in a series configuration. Out of these, one acts as the “master” and the other as a “slave”. The output from the master flip flop is connected to the two inputs of the slave flip flop whose output is fed back to inputs of the master flip flop.

In addition to these two flip-flops, the circuit also includes an inverter. The inverter is connected to clock pulse in such a way that the inverted clock pulse is given to the slave flip-flop. In other words if $CP=0$ for a master flip-flop, then $CP=1$ for a slave flip-flop and if $CP=1$ for master flip flop then it becomes 0 for slave flip flop.

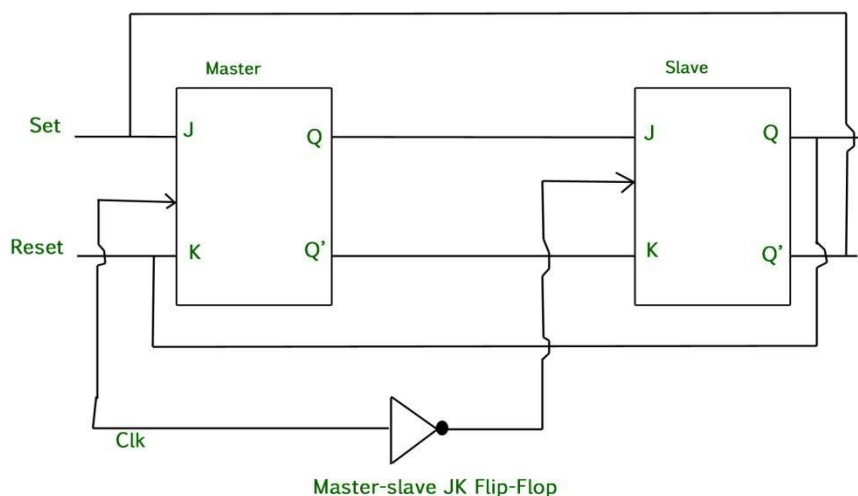


Fig.3.2.1 Master Slave JK flip flop.

Working of a master slave flip flop –

When the clock pulse goes to 1, the slave is isolated; J and K inputs may affect the state of the system. The slave flip-flop is isolated until the CP goes to 0. When the CP goes back to 0, information is passed from the master flip-flop to the slave and output is obtained.

Firstly the master flip flop is positive level triggered and the slave flip flop is negative level triggered, so the master responds before the slave.

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If $J=0$ and $K=1$, the high Q' output of the master goes to the K input of the slave and the clock forces the slave to reset, thus the slave copies the master.

If $J=1$ and $K=0$, the high Q output of the master goes to the J input of the slave and the Negative transition of the clock sets the slave, copying the master.

If $J=1$ and $K=1$, it toggles on the positive transition of the clock and thus the slave toggles on the negative transition of the clock.

If $J=0$ and $K=0$, the flip flop is disabled and Q remains unchanged.

Let's check the take away from this lecture

1) Which of the following flip-flops is free from the race around the problem?

- a) **T flip-flop**
- b) SR flip-flop
- c) Master-Slave Flip-flop
- d) D flip-flop

2) Master slave flip flop is also referred to as?

- a) Level triggered flip flop
- b) **Pulse triggered flip flop**
- c) Edge triggered flip flop
- d) Edge-Level triggered flip flop

3) On a master-slave flip-flop, when is the master enabled?

- a) when the gate is LOW
- b) **when the gate is HIGH**
- c) both of the above
- d) neither of the above

Learning from this lecture: Learners will be able to understand the concept of Race around condition and Master Slave J-K Flip Flops

Lecture : 19

Timing Diagram, Flip-flop conversion

Timing Diagram of a Master flip flop

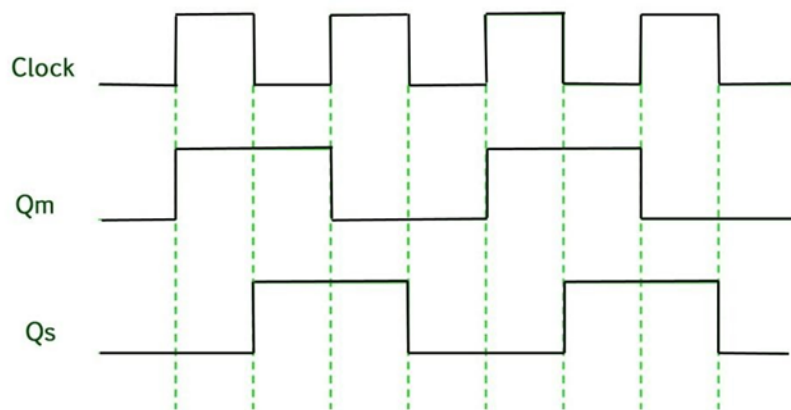


Fig.3.3.1 Timing Diagram of a Master flip flop.

1. When the Clock pulse is high the output of master is high and remains high till the clock is low because the state is stored.
2. Now the output of master becomes low when the clock pulse becomes high again and remains low until the clock becomes high again.
3. Thus toggling takes place for a clock cycle.
4. When the clock pulse is high, the master is operational but not the slave thus the output of the slave remains low till the clock remains high.
5. When the clock is low, the slave becomes operational and remains high until the clock again becomes low.
6. Toggling takes place during the whole process since the output is changing once in a cycle.

This makes the Master-Slave J-K flip flop a Synchronous device as it only passes data with the timing of the clock signal.

Flip-flop conversion

For the conversion of one flip flop to another, a combinational circuit has to be designed first. If a JK Flip Flop is required, the inputs are given to the combinational circuit and the output of the combinational circuit is connected to the inputs of the actual flip flop. Thus, the output of the actual flip flop is the output of the required flip flop. In this section, the following flip flop conversions will be explained.

SR Flip Flop to JK Flip Flop

JK Flip Flop to SR Flip Flop

SR Flip Flop to D Flip Flop

D Flip Flop to SR Flip Flop

JK Flip Flop to T Flip Flop

JK Flip Flop to D Flip Flop

D Flip Flop to JK Flip Flop

SR Flip Flop to JK Flip Flop

As told earlier, J and K will be given as external inputs to S and R. As shown in the logic diagram below, S and R will be the outputs of the combinational circuit.

The truth tables for the flip flop conversion are given below. The present state is represented by Q_p and Q_{p+1} is the next state to be obtained when the J and K inputs are applied.

For two inputs J and K, there will be eight possible combinations. For each combination of J, K and Q_p , the corresponding Q_{p+1} states are found. Q_{p+1} simply suggests the future values to be obtained by the JK flip flop after the value of Q_p . The table is then completed by writing the values of S and R required to get each Q_{p+1} from the corresponding Q_p . That is, the values of S and R that are required to change the state of the flip flop from Q_p to Q_{p+1} are written.

S-R Flip Flop to J-K Flip Flop

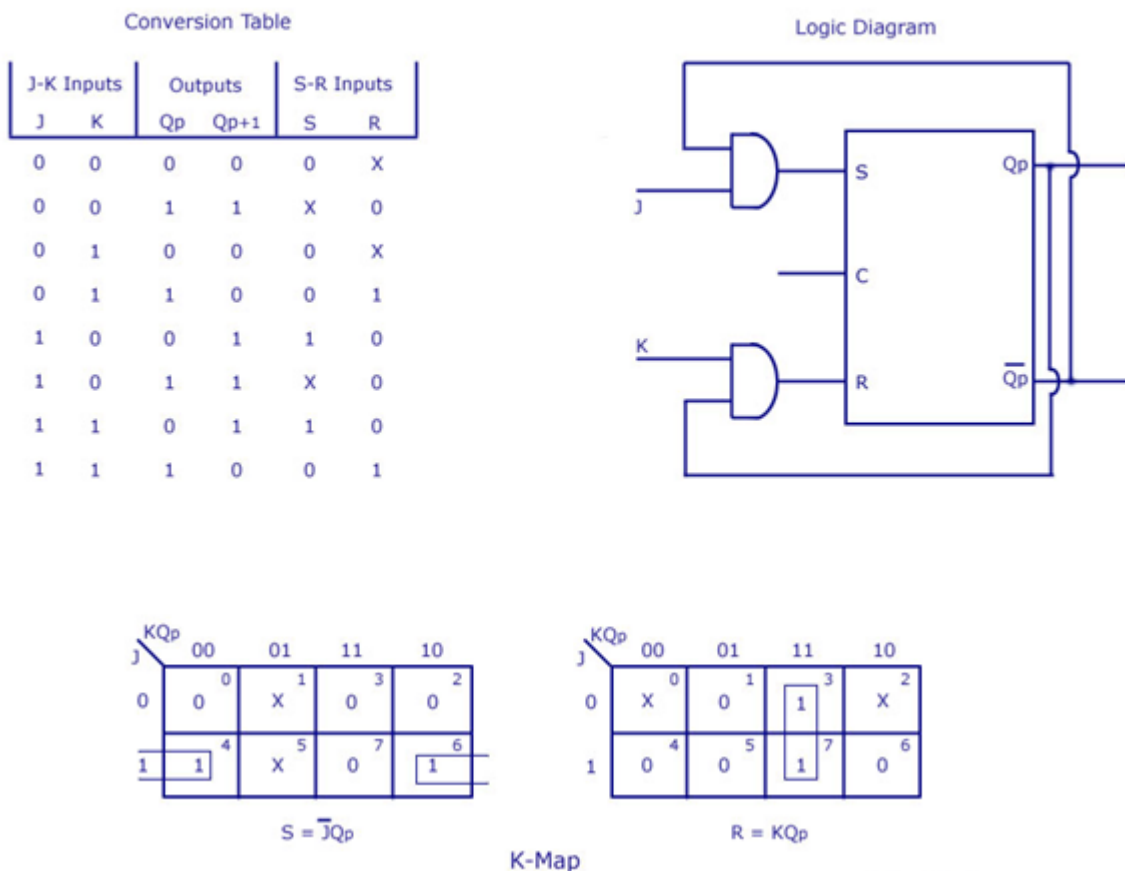


Fig.3.3.2 Conversion from SR to JK flip flop.

JK Flip Flop to SR Flip Flop

This will be the reverse process of the above explained conversion. S and R will be the external inputs to J and K. As shown in the logic diagram below, J and K will be the outputs of the combinational circuit. Thus, the values of J and K have to be obtained in terms of S, R and Q_p. The logic diagram is shown below.

A conversion table is to be written using S, R, Q_p, Q_{p+1}, J and K. For two inputs, S and R, eight combinations are made. For each combination, the corresponding Q_{p+1} outputs are found out. The outputs for the combinations of S=1 and R=1 are not permitted for an SR flip flop. Thus the outputs are considered invalid and the J and K values are taken as “don’t cares”.

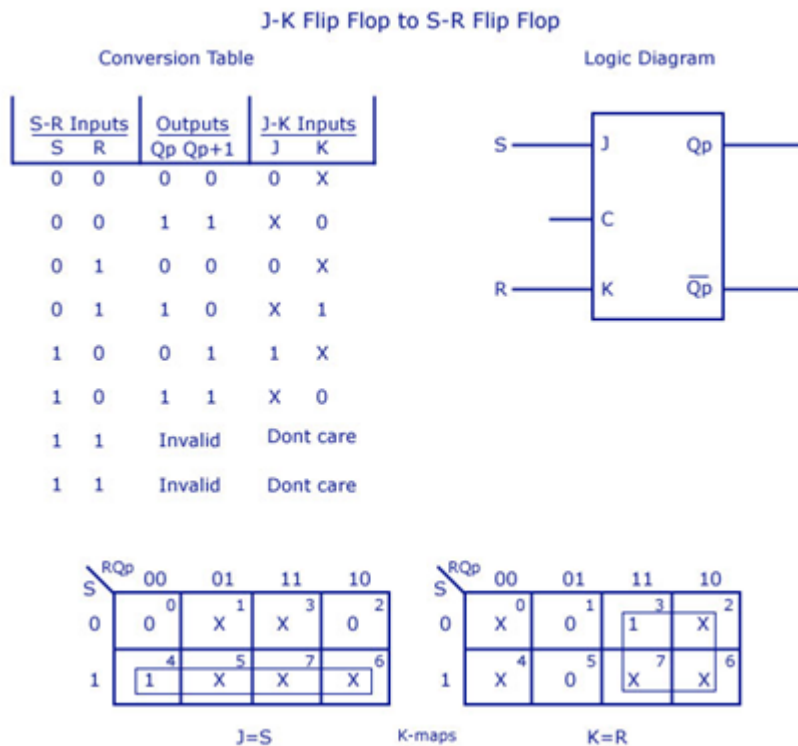


Fig.3.3.3 Conversion from JK to SR flip flop.

SR Flip Flop to D Flip Flop

As shown in the figure, S and R are the actual inputs of the flip flop and D is the external input of the flip flop. The four combinations, the logic diagram, conversion table, and the K-map for S and R in terms of D and Qp are shown below.

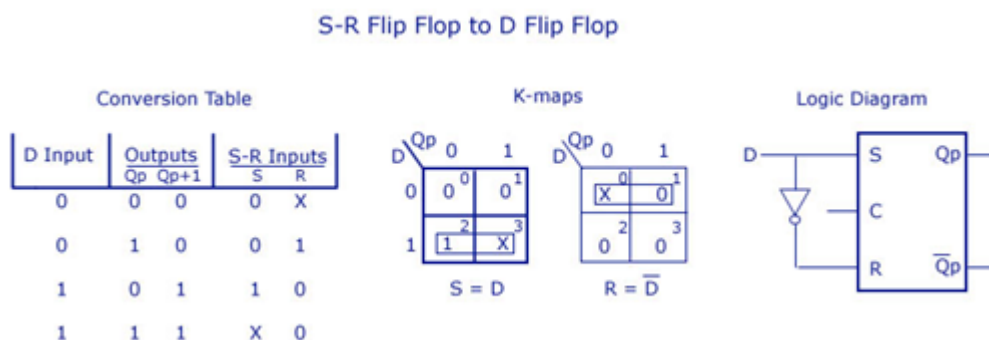


Fig.3.3.4 Conversion from SR to D flip flop.

D Flip Flop to SR Flip Flop

D is the actual input of the flip flop and S and R are the external inputs. Eight possible combinations are achieved from the external inputs S, R and Qp. But, since the combination

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of $S=1$ and $R=1$ are invalid, the values of Q_{p+1} and D are considered as “don’t cares”. The logic diagram showing the conversion from D to SR , and the K-map for D in terms of S , R and Q_p are shown below.

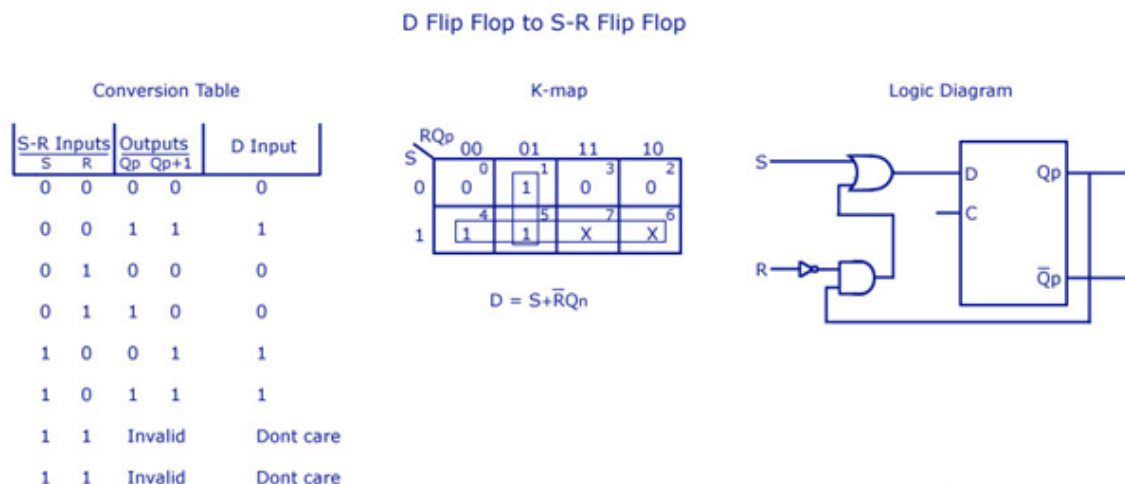


Fig.3.3.5 Conversion from D to SR flip flop.

JK Flip Flop to T Flip Flop

J and K are the actual inputs of the flip flop and T is taken as the external input for conversion. Four combinations are produced with T and Q_p . J and K are expressed in terms of T and Q_p . The conversion table, K-maps, and the logic diagram are given below.

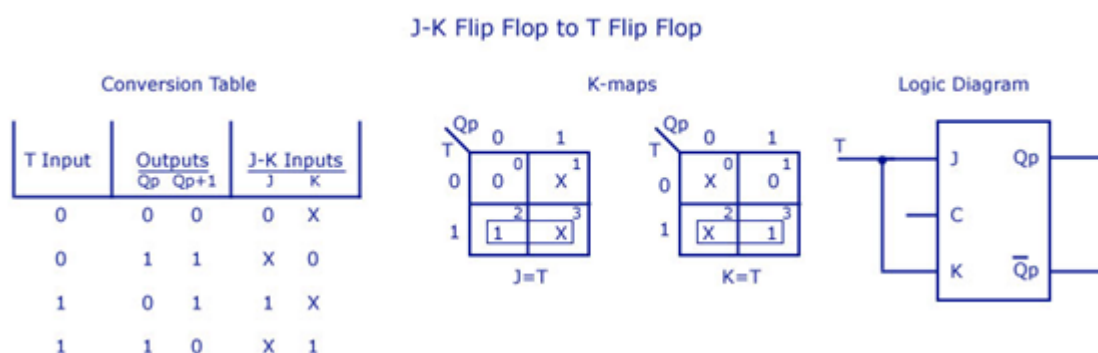


Fig.3.3.6 Conversion from JK to T flip flop.

JK Flip Flop to D Flip Flop

D is the external input and J and K are the actual inputs of the flip flop. D and Q_p make four combinations. J and K are expressed in terms of D and Q_p . The four combination conversion table, the K-maps for J and K in terms of D and Q_p , and the logic diagram showing the conversion from JK to D are given below.

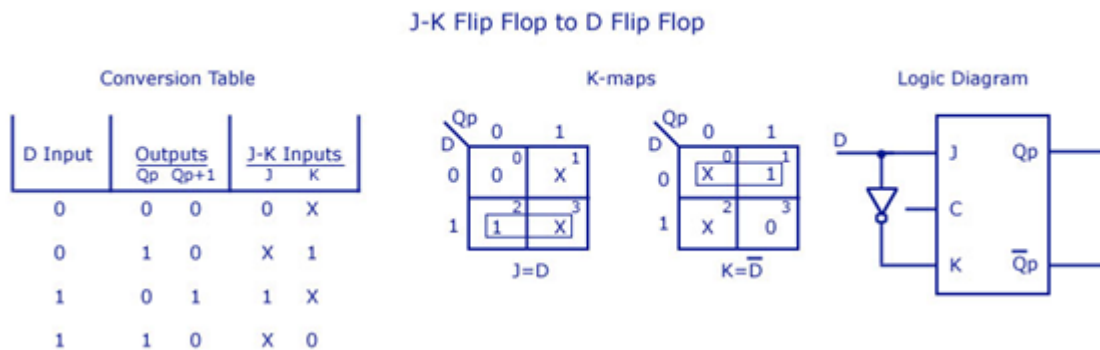


Fig.3.3.7 Conversion from JK to D flip flop.

D Flip Flop to JK Flip Flop

In this conversion, D is the actual input to the flip flop and J and K are the external inputs. J, K and Q_p make eight possible combinations, as shown in the conversion table below. D is expressed in terms of J, K and Q_p .

The conversion table, the K-map for D in terms of J, K and Q_p and the logic diagram showing the conversion from D to JK are given in the figure below.

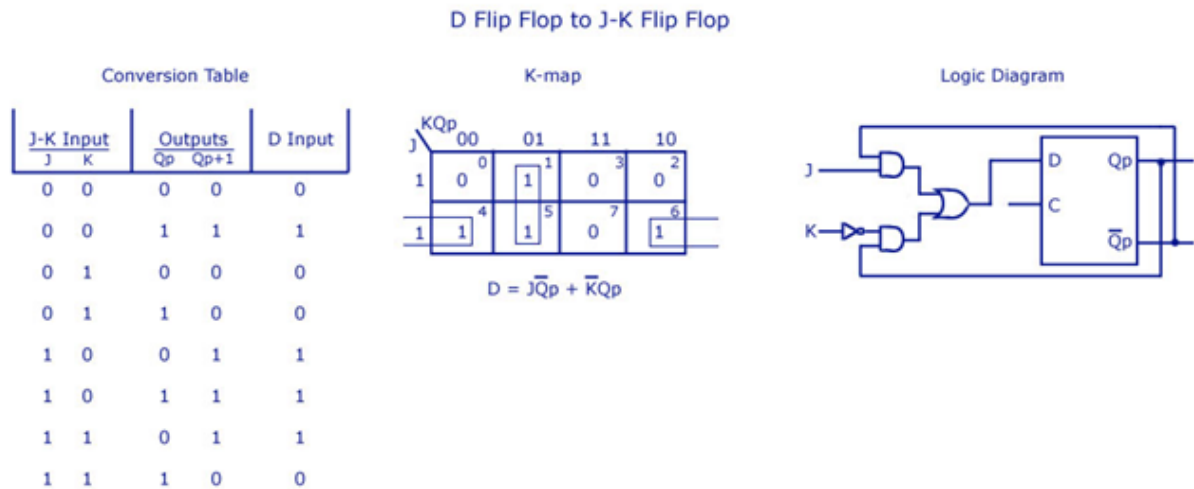


Fig.3.3.8 Conversion from D to JK flip flop.

Let's check the take away from this lecture

- 1) Which of the following describes the operation of a positive edge-triggered D flip-flop?
 - a) If both inputs are HIGH, the output will toggle.
 - b) The output will follow the input on the leading edge of the clock.**
 - c) When both inputs are LOW, an invalid state exists.
 - d) The input is toggled into the flip-flop on the leading edge of the clock and is passed to the output on the trailing edge of the clock.

- 2) On a positive edge-triggered S-R flip-flop, the outputs reflect the input condition when
 - a) the clock pulse is LOW
 - b) the clock pulse is HIGH
 - c) the clock pulse transitions from LOW to HIGH**
 - d) the clock pulse transitions from HIGH to LOW

Learning from this lecture: Learners will be able to understand the concept of Timing Diagram and Flip-flop conversion.

Lecture : 20

Introduction to Counters: Design of Asynchronous and Synchronous Counters, Modulus of the Counters

Counters are sequential logic circuits that proceed through a well defined sequence of states after application of clock pulses.

- Special type of registers with a capability of counting with the application of clock pulse
- Counters are used for a counting pulses
- Counters are constructed using Flipflops and logic gates
- Counters are classified into two categories
 1. Ripple (or Asynchronous) Counters
 2. Synchronous Counters

Asynchronous or ripple counters

The logic diagram of a 2-bit ripple up counter is shown in figure. The toggle (T) flip-flop are being used. But we can use the JK flip-flop also with J and K connected permanently to logic 1. External clock is applied to the clock input of flip-flop A and QA output is applied to the clock input of the next flip-flop i.e. FF-B.

Logical Diagram

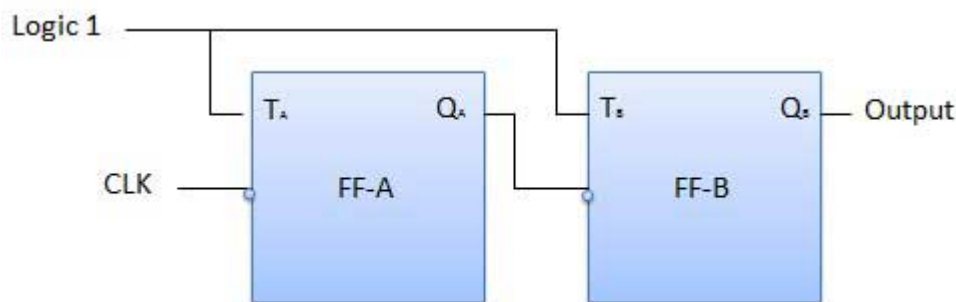


Fig.3.4.1 Logical Diagram of Asynchronous or ripple counters.

Operation

1. Initially let both the FFs be in the reset state

$QBQA = 00$ initially

2. After 1st negative clock edge

- As soon as the first negative clock edge is applied, FF-A will toggle and QA will be equal to 1.
- QA is connected to clock input of FF-B. Since QA has changed from 0 to 1, it is treated as the positive clock edge by FF-B. There is no change in QB because FF-B is a negative edge triggered FF.
- QBQA = 01 after the first clock pulse.

3. After 2nd negative clock edge

- On the arrival of second negative clock edge, FF-A toggles again and QA = 0.
- The change in QA acts as a negative clock edge for FF-B. So it will also toggle, and QB will be 1.
- QBQA = 10 after the second clock pulse.

4. After 3rd negative clock edge

- On the arrival of 3rd negative clock edge, FF-A toggles again and QA becomes 1 from 0.
- Since this is a positive going change, FF-B does not respond to it and remains inactive. So QB does not change and continues to be equal to 1.
- QBQA = 11 after the third clock pulse.

5. After 4th negative clock edge

- On the arrival of 4th negative clock edge, FF-A toggles again and QA becomes 1 from 0.
- This negative change in QA acts as clock pulse for FF-B. Hence it toggles to change QB from 1 to 0.
- QBQA = 00 after the fourth clock pulse.

Truth Table

Clock	Counter output		State number	Deciimal Counter output
	Q _a	Q _b		
Initially	0	0	—	0
1st	0	1	1	1
2nd	1	0	2	2
3rd	1	1	3	3
4th	0	0	4	0

Fig.3.4.2 Truth Table of Asynchronous or ripple counters.

Synchronous counters

If the "clock" pulses are applied to all the flip-flops in a counter simultaneously, then such a counter is called as synchronous counter.

2-bit Synchronous up counter

The JA and KA inputs of FF-A are tied to logic 1. So FF-A will work as a toggle flip-flop. The JB and KB inputs are connected to QA.

Logical Diagram

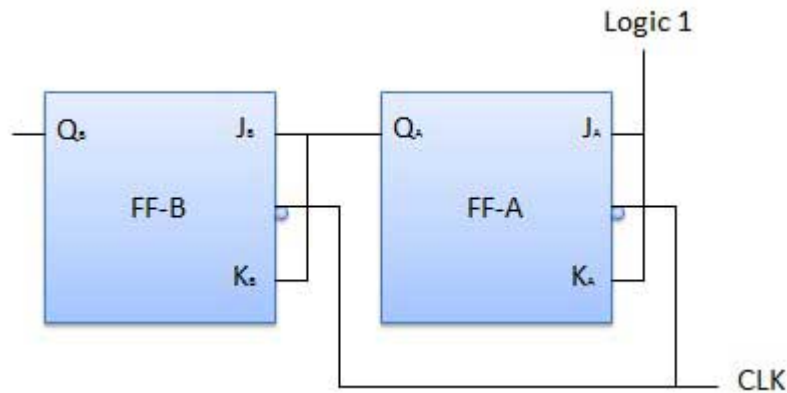


Fig.3.4.3 Logical Diagram of Synchronous counters.

Operation

1.Initially let both the FFs be in the reset state

$QBQA = 00$ initially.

2.After 1st negative clock edge

- As soon as the first negative clock edge is applied, FF-A will toggle and QA will change from 0 to 1.
- But at the instant of application of negative clock edge, QA , JB = KB = 0. Hence FF-B will not change its state. So QB will remain 0.
- $QBQA = 01$ after the first clock pulse.

3.After 2nd negative clock edge

- On the arrival of second negative clock edge, FF-A toggles again and QA changes from 1 to 0.
- But at this instant QA was 1. So JB = KB= 1 and FF-B will toggle. Hence QB changes from 0 to 1.
- $QBQA = 10$ after the second clock pulse.

4.After 3rd negative clock edge

Module 3 Sequential Logic Design

- On application of the third falling clock edge, FF-A will toggle from 0 to 1 but there is no change of state for FF-B.
- QBQA = 11 after the third clock pulse.

5. After 4th negative clock edge

- On application of the next clock pulse, QA will change from 1 to 0 as QB will also change from 1 to 0.
- QBQA = 00 after the fourth clock pulse.

Modulus Counter

Modulus Counters, or simply MOD counters, are defined based on the number of states that the counter will sequence through before returning back to its original value. For example, a 2-bit counter that counts from 00 to 11 in binary, that is 0 to 3 in decimal, has a modulus value of 4 (00 → 01 → 10 → 11, and return back to 00) so would therefore be called a modulo-4, or mod-4, counter. Note also that it has taken four clock pulses to get from 00 to 11.

As in this simple example there are only two bits, ($n = 2$) then the maximum number of possible output states (maximum modulus) for the counter is: $2^n = 2^2$ or 4. However, counters can be designed to count to any number of 2^n states in their sequence by cascading together multiple counting stages to produce a single modulus or MOD-N counter.

Therefore, a “Mod-N” counter will require “N” number of flip-flops connected together to count a single data bit while providing 2^n different output states, (n is the number of bits). Note that N is always a whole integer value.

The 2-bit ripple counter is called as MOD-4 counter and 3-bit ripple counter is called as MOD-8 counter. So in general, an n-bit ripple counter is called as modulo-N counter. Where, MOD number = 2^n .

Type of modulus

2-bit up or down (MOD-4)

3-bit up or down (MOD-8)

4-bit up or down (MOD-16)

Let's check the take away from this lecture

1. In digital logic, a counter is a device which _____

a) Counts the number of outputs

b) Stores the number of times a particular event or process has occurred

c) Stores the number of times a clock pulse rises and falls

d) Counts the number of inputs

2. Ripple counters are also called _____

a) SSI counters

b) Asynchronous counters

c) Synchronous counters

d) VLSI counters

3. Internal propagation delay of asynchronous counter is removed by _____

a) Ripple counter

b) Ring counter

c) Modulus counter

d) Synchronous counter

Exercise

Q.1 Design Asynchronous Counter.

Q.2 Design Synchronous Counter.

Q.3 Explain Modulus of the Counters in detail.

Learning from this lecture: Learners will be able to understand the concept of Counters

Lecture : 21

UP- DOWN counter, Ring and twisted ring/Johnson Counter, sequence generator

Ring counter is a sequential logic circuit that is constructed using shift register. Same data recirculates in the counter depending on the clock pulse.

Ring counters are of two types

1) Ordinary Ring counters

2) Johnson counter

4 bit Ring Counter

The ring counter is a cascaded connection of flip flops, in which the output of last flip flop is connected to input of first flip flop. In ring counter if the output of any stage is 1, then its remainder is 0. The Ring counters transfers the same output throughout the circuit.

That means if the output of the first flip flop is 1, then this is transferred to its next stage i.e. 2nd flip flop. By transferring the output to its next stage, the output of first flip flop becomes 0. And this process continues for all the stages of a ring counter. If we use n flip flops in the ring counter, the '1' is circulated for every n clock cycles.

The circuit diagram of the ring counter

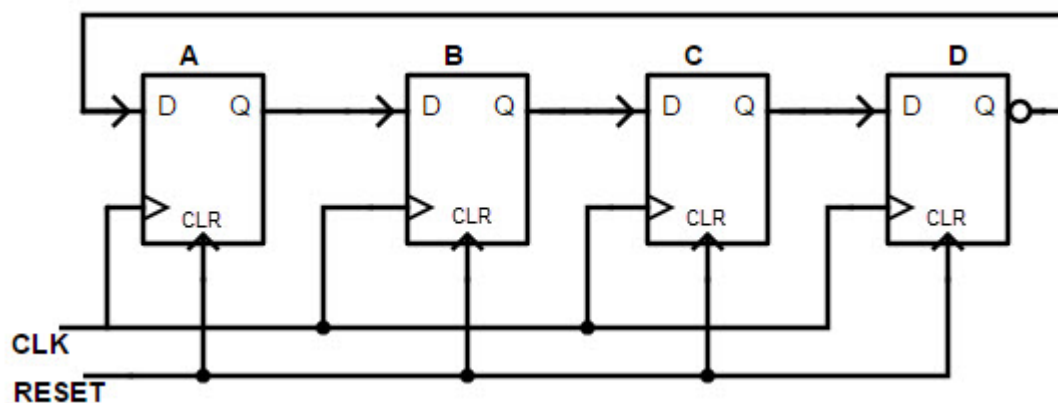


Fig.3.5.1 The circuit diagram of the ring counter

Here we design the ring counter by using D flip flop. This is a Mod 4 ring counter which has 4 D flip flops connected in series. The clock signal is applied to clock input of each flip flop, simultaneously and the RESET pulse is applied to the CLR inputs of all the flip flops.

Operation of Ring Counter

Initially, all the flip flops in ring counter are reset to 0 by applying CLEAR signal. Before applying the clock pulse, we apply the PRESET pulse to the flip flops which assigns the value '1' to the ring counter circuit. For each clock signal, the data circulates among all the 4 flip flop stages of ring counter.

This 4 staged ring counter is called Mod 4 ring counter or 4 bit ring counter. To circulate the data correctly in the ring counter, we must load the counter with required values like all 0's or all 1's.

Truth table of ring counter

The truth table of the 4 bit ring counter

Q_0	Q_1	Q_2	Q_3
1	0	0	0
0	1	0	0
0	0	1	0
0	0	0	1

Fig.3.5.2 The truth table of the 4 bit ring counter.

When CLEAR input $CLR = 0$, then all flip flops are set to 1. When CLEAR input $CLR = 1$, the ring counter starts its operation. For one clock signal, the counter starts its operation. On next clock signal, the counter again resets to 0000. Ring counter has 4 sequences: 0001, 0010, 0100, 1000, 000.

Timing diagram of Ring Counter

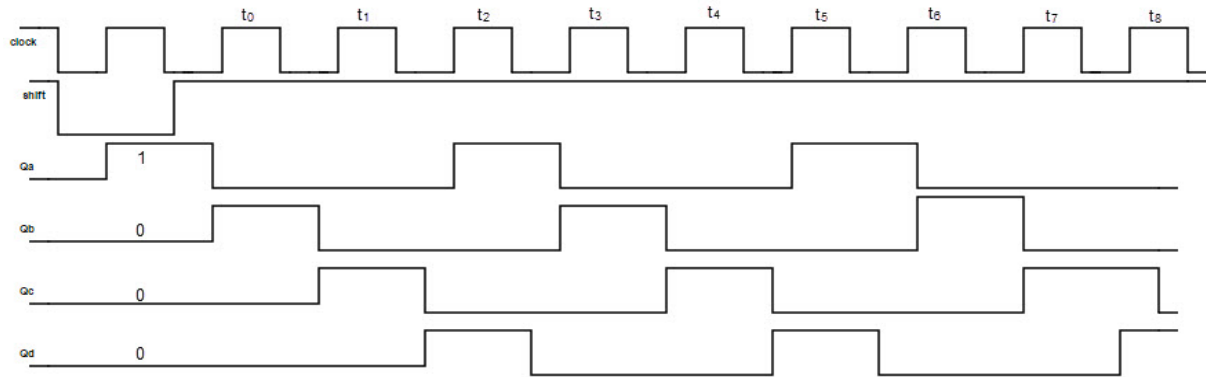


Fig.3.5.3 Timing diagram of Ring Counter

The timing diagram of the Ring counter will explain that the clock signal changes the output of every stage of the counter, so that CLK signal will help the data to circulate from one flip flop to another. As the 4 bit ring counter (4 stages or 4 flip flops) circulates the preset digit within one clock signal, the output frequency of each flip flop is $\frac{1}{4}$ th of the main clock frequency.

State diagram of ring counter

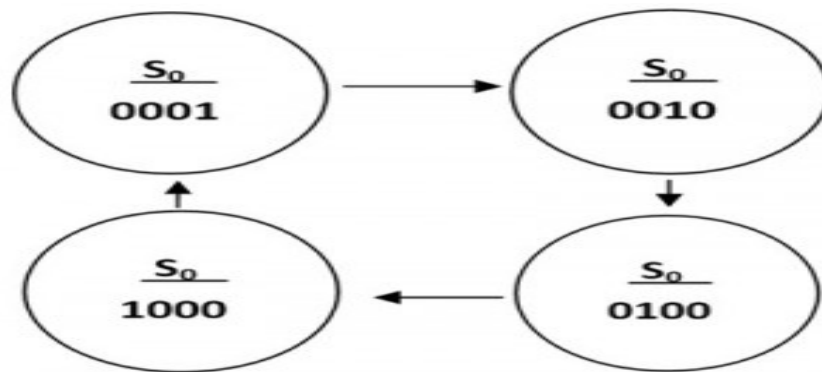


Fig.3.5.4 State diagram of Ring Counter.

The state diagram of the 4 bit ring counter is shown in above picture. It denotes that the position of the preset digit (in this case preset digit is 1) is changing its position from LSB to MSB, for one clock signal.

Advantages

Module 3 Sequential Logic Design

- Can be implemented using D and JK flip-flops. It is a self-decoding circuit.

Disadvantages

- Only four of the 15 states are being utilized.

Johnson Counter

The Johnson counter is a modification of ring counter. In this the inverted output of the last stage flip flop is connected to the input of first flip flop. If we use n flip flops to design the Johnson counter, it is known as $2n$ bit Johnson counter or Mod $2n$ Johnson counter.

This is an advantage of the Johnson counter that it requires only half number of flip flops that of a ring counter uses, to design the same Mod.

The main difference between the 4 bit ring counter and the Johnson counter is that, in ring counter, we connect the output of last flip flop directly to the input of first flip flop. But in Johnson counter, we connect the inverted output of last stage to the first stage input.

The Johnson counter is also known as Twisted Ring Counter, with a feedback. In Johnson counter the input of the first flip flop is connected from the inverted output of the last flip flop.

The Johnson counter or switch trail ring counter is designed in such a way that it overcomes the limitations of ring counter. Mainly it reduces the number of flip flops required for designing the circuit.

Q_A	Q_B	Q_C	Q_D
0	0	0	0
1	0	0	0
1	1	0	0
1	1	1	0
1	1	1	1
0	1	1	1
0	0	1	1
0	0	0	1
repeat			

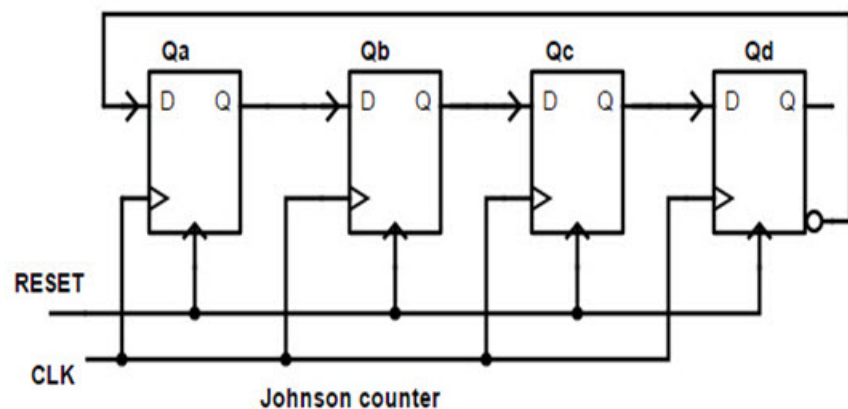


Fig.3.5.5 Circuit diagram and Truth table of Johnson counter.

Module 3 Sequential Logic Design

Similar to the ring counter, the clock signal in johnson counter is connected to the clock input of each flip flop simultaneously.

Operation of johnson counter

The Johnson counter designed with D flip flop is shown below. It has four stages i.e. four flip flops connected in series type or cascaded. Initially zero / Null is fed to the Johnson counter and on applying the clock signal, outputs will change to “1000”, “1100”, “1110”, “1111”, “0111”, “0011”, “0001”, “0000” in a sequence and the sequence will repeat for next clock signal.

The Johnson counter produces a special pattern by passing four 0's and then four 1's and thus it produces a special pattern by counting up then down.

Truth table of johnson counter

The truth table of the 4 bit ring counter

Q _A	Q _B	Q _C	Q _D
0	0	0	0
1	0	0	0
1	1	0	0
1	1	1	0
1	1	1	1
0	1	1	1
0	0	1	1
0	0	0	1
repeat			

Fig.3.5.6 The truth table of the 4 bit ring counter.

The state diagram indicates that how the data transfers from one flip flop to another for every clock pulse. The 4 stage Johnson ring counters are used as frequency dividers, by varying their feedback connections. So they can be used as frequency divider circuits also.

Timing diagram of johnson counter

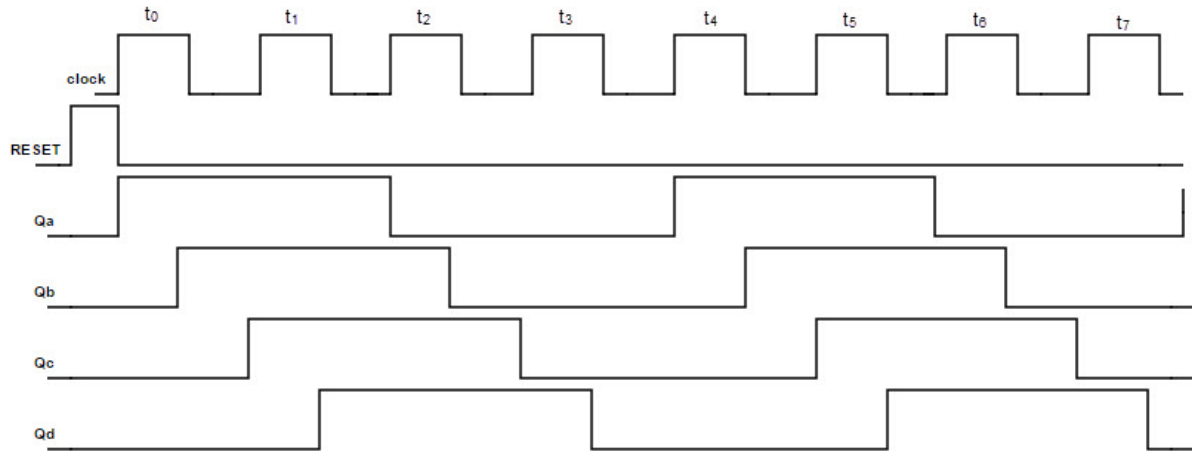


Fig.3.5.7 Timing diagram of Johnson counter.

The timing diagram of the johnson counter will explain that the clock signal changes the output of every stage of the counter, so that CLK signal will help the data to circulate from one flip flop to another.

Applications of Ring counters

- Ring counters are used to count the data in a continuous loop.
- They are also used to detect the various numbers values or various patterns within a set of information, by connecting AND & OR logic gates to the ring counter circuits.
- 2 stage, 3 stage and 4 stage ring counters are used in frequency divider circuits as divide by 2 and divide by 3 and divide by 4 circuits, respectively.
- The 3 stage Johnson counter is used as a 3 phase square wave generator which produces 120° phase shift.
- The 5 stage Johnson counter circuit is generally used as synchronous decade (BCD) counter and also as divider circuit.
- The 2 stage Johnson counters are also known as “Quadrature oscillator” which is used to produce 4 level individual outputs which are out of phase with 90° with each other. This quadrature generator is used to produce 4 phase timing signal.

A sequence generator is one kind of digital logic circuit. The main function of this is to generate a set of outputs. Every output is one of a number of binary or Q-ary logic levels or symbols. The length of the series may be indefinite otherwise fixed. A special kind of sequence generator is a binary counter. These generators are utilized in a wide variety of applications like coding & control.

The sequence generator circuit is used to generate a prescribed series of bits in synchronization through a CLK. This kind of generator is used as a code generator, counters,

random bit generators, sequence, and prescribed period generator. The basic design diagram of this is shown below.

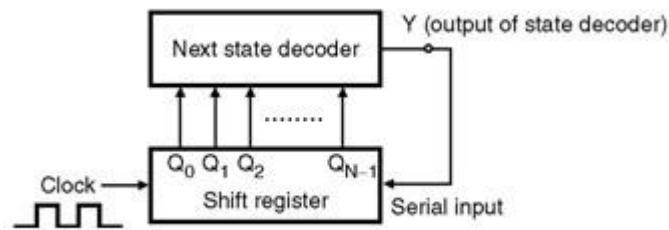


Fig.3.5.7 Sequence Generator Structure

The N-bit shift register outputs like Q_0 through Q_{N-1} are applied like the inputs to a combinational circuit is known as the next state decoder. Here, the output of a next state decoder 'Y' is given as the serial input to the shift register. The designing of the next state decoder is done based on the sequence required.

Sequence Generator using Counters

The sequence generator block diagram using a counter is illustrated below. Here, the combinational circuit is the next state decoder. The input of this state decoder can be obtained from the outputs of the FFs. Similarly, the outputs of this state decoder are given as inputs to the flip-flops. Based on the number of FFs, the required sequence like 0's or 1's can be given and this can be generated like 1011011.

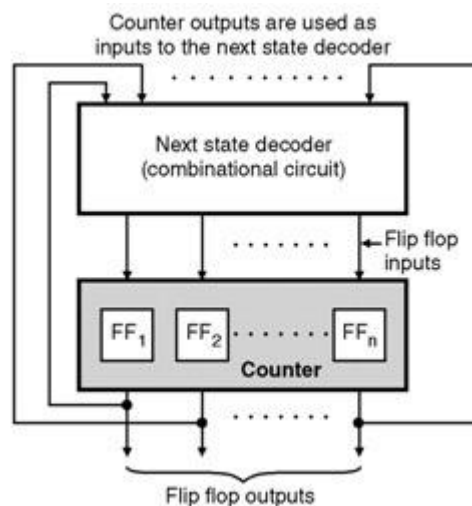


Fig.3.5.8 Sequence Generator using Counter

Module 3 Sequential Logic Design

The number of flip flops can be decided through the given sequence like the following.

- First, count the number of zeros and ones in the given sequence.
- Select the high number of the two. And let this number will be 'N'.
- The no. of flip flops can be calculated as $N = 2^{n-1}$
- For instance, the given sequence is 1011011, where the number of ones is 5 and the number of zeros is two. So choose a higher one from them that is 5. So $5 = 2^{n-1}$, so $n=4$ FFs will be necessary.

Properties

The sequence generator properties include the following.

- Use Shared Sequence
- Reset
- Increment By
- Number of Cached Values
- End Value
- Cycle Start Value
- Initial Value
- Cycle

Steps Involved in Designing Sequence Generator using D Flip-Flops

We know the function of a counter that allows an exact number of states in a prearranged sequence. For instance, an up-counter with 3-bit counts 0 to 7 whereas a similar order is upturned in the case of down counter.

There are different ways to design the circuits can using FFs, multiplexers. Here we are designing a sequence generator using D FFs in different steps. Similarly, there are different steps involved in designing a sequence generator using JK Flip-Flops.

Let's take an example that we aim to design a circuit that moves throughout the states of 0-1-3-2 before doing again the similar pattern. The steps involved throughout this method are as follows.

In Step-1

Firstly, we need to decide the no. of FFs which would be necessary to get our object. In the following example, there are four states which are equal to the 2-bit counter states excluding the order where they transfer. From this, one can estimate the necessity of FFs to be two in order to attain our object.

In Step-2

From the step1, let's design the state transition table for our sequence generator which is illustrated through the initial four columns in the table. In that, the primary two columns

specify the present states and the next states. For example, in the first state of our example is “0 = 00” so it leads to the second state that is next state 1 = “01”.

In Step-3

In the state transition table is extended by including the excitation table of the FFs. In this case, the excitation table of the D flip-flop is the fifth & the sixth columns of the table. For instance, look at the present and next states in the table like 1 & 0 respectively then it results ‘0’ in D1. In the following table, the first two columns represent the present state, the second two columns represent the next states and the last two are inputs of D-FF.

Q1	Q0	Q1+	Q0+	D1	D0
0	0	0	1	0	1
0	1	1	1	1	1
1	1	1	0	1	0
1	0	0	0	0	0

Fig.3.5.9 Sequence Generator using D flip flop state transition table

In Step-4

In this step, the Boolean expressions for D0 & D1 can be derived with the help of a K-map. But this example is quite easy; so by using Boolean laws, we can solve for D1 & D0. Therefore

$$D0 = Q1'Q0' + Q1' Q0 = Q1' (Q0' + Q0) = Q1'(1) = Q1'$$

$$D1 = Q1'Q0 + Q1 Q0 = Q0 (Q1' + Q1) = Q0 (1) = Q0$$

In Step-5

The sequence generator can be designed using the D FFs based on inputs like the following.

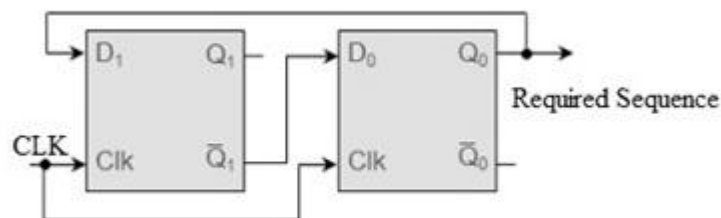


Fig.3.5.10 Sequence Generator using D-FFs

In the above circuit, the preferred series is generated depending on the supplied CLK pulses. So it must be noted that the similarity existing here for an easy design can be successfully extended to produce a longer series of bits.

Let's check the take away from this lecture

1. Ring shift and Johnson counters are _____

a) **Synchronous counters**

b) Asynchronous counters

c) True binary counters

d) Synchronous and true binary counters

2. Which sequential circuits are applicable for counting pulses?

a. **Counters**

b. Flip Flops

c. Registers

d. Latches

Exercise

Q.1 Explain UP- DOWN counter.

Q.2 Explain Ring and twisted ring/Johnson Counter.

Q.3 Explain sequence generator.

Learning from this lecture: Learners will be able to understand the concept of UP- DOWN counter, Ring and twisted ring/Johnson Counter, and sequence generator.

Lecture : 22

Introduction to Registers: SISO, SIPO, PIPO, PISO

Flip-flop is a 1 bit memory cell which can be used for storing the digital data. To increase the storage capacity in terms of number of bits, we have to use a group of flip-flop. Such a group of flip-flop is known as a Register. The n-bit register will consist of n number of flip-flop and it is capable of storing an n-bit word.

Module 3 Sequential Logic Design

The binary data in a register can be moved within the register from one flip-flop to another. The registers that allow such data transfers are called as shift registers. There are four mode of operations of a shift register.

- Serial Input Serial Output
- Serial Input Parallel Output
- Parallel Input Parallel Output
- Parallel Input Serial Output

Serial Input Serial Output

Let all the flip-flop be initially in the reset condition i.e. $Q_3 = Q_2 = Q_1 = Q_0 = 0$. If an entry of a four bit binary number 1 1 1 1 is made into the register, this number should be applied to Din bit with the LSB bit applied first. The D input of FF-3 i.e. D₃ is connected to serial data input Din. Output of FF-3 i.e. Q₃ is connected to the input of the next flip-flop i.e. D₂ and so on.

Block Diagram

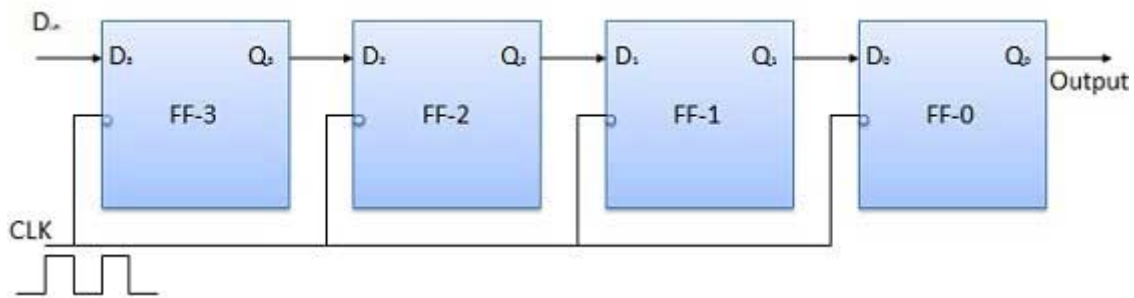


Fig.3.6.1 Block Diagram of Serial Input Serial Output

Operation

Before application of clock signal, let $Q_3 Q_2 Q_1 Q_0 = 0000$ and apply LSB bit of the number to be entered to Din. So $D_{in} = D_3 = 1$. Apply the clock. On the first falling edge of clock, the FF-3 is set, and stored word in the register is $Q_3 Q_2 Q_1 Q_0 = 1000$.

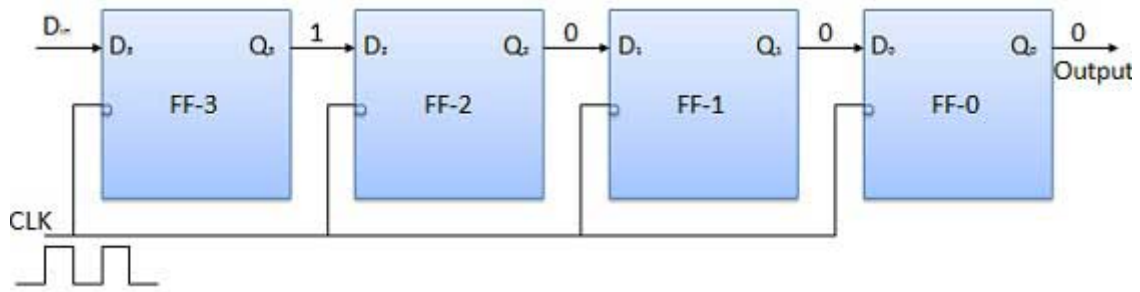


Fig.3.6.2 Operation 1 of Serial Input Serial Output

Apply the next bit to Din. So $D_{in} = 1$. As soon as the next negative edge of the clock hits, FF-2 will set and the stored word change to $Q_3 Q_2 Q_1 Q_0 = 1100$.

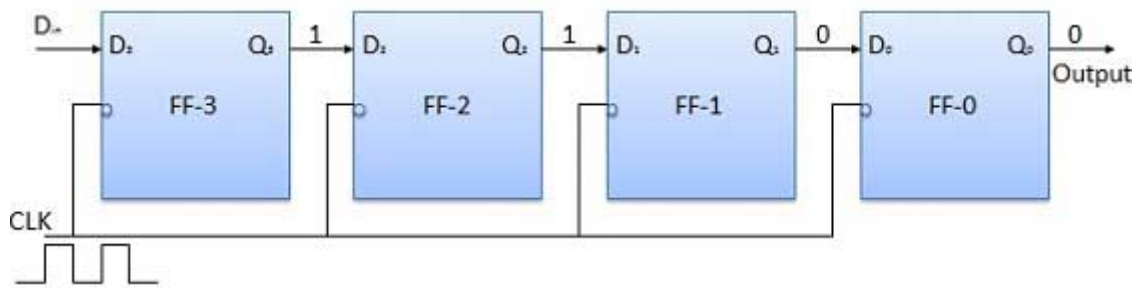


Fig.3.6.3 Operation 2 of Serial Input Serial Output

Similarly with $D_{in} = 1$ and with the fourth negative clock edge arriving, the stored word in the register is $Q_3 Q_2 Q_1 Q_0 = 1111$.

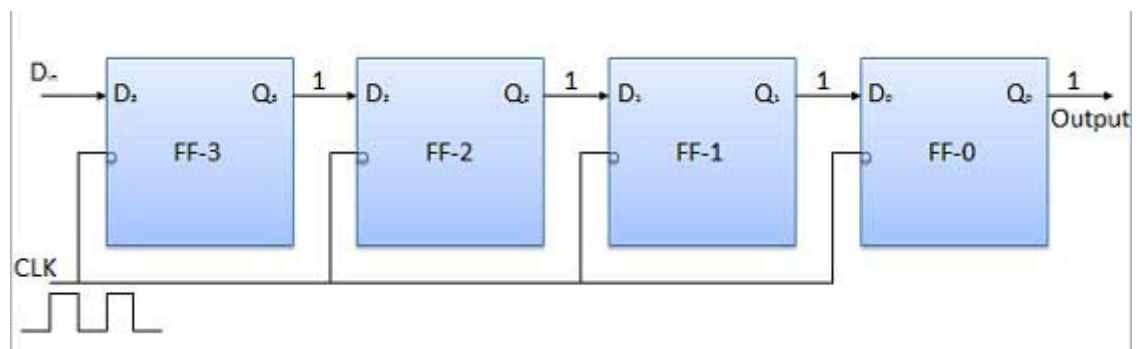


Fig.3.6.4 Operation 3 of Serial Input Serial Output

Truth Table

	CLK	$D_n = Q_3$	$Q_3 = D_2$	$Q_2 = D_1$	$Q_1 = D_0$	Q_0
Initially			0	0	0	0
(i)	↓	1	1	0	0	0
(ii)	↓	1	1	1	0	0
(iii)	↓	1	1	1	1	0
(iv)	↓	1	1	1	1	1

→ Direction of data travel

Fig.3.6.5 Truth Table of Serial Input Serial Output

Waveforms



Fig.3.6.6 Waveforms of Serial Input Serial Output

Serial Input Parallel Output

In such types of operations, the data is entered serially and taken out in parallel fashion.

Data is loaded bit by bit. The outputs are disabled as long as the data is loading.

As soon as the data loading gets completed, all the flip-flops contain their required data, the outputs are enabled so that all the loaded data is made available over all the output lines at the same time.

4 clock cycles are required to load a four bit word. Hence the speed of operation of SIPO mode is same as that of SISO mode.

Block Diagram

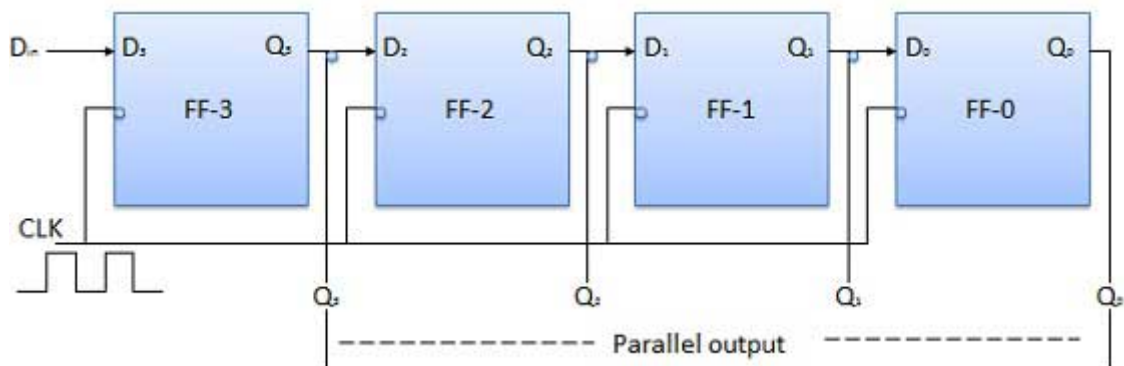


Fig.3.6.7 Block Diagram of Serial Input Parallel Output

Parallel Input Parallel Output (PIPO)

In this mode, the 4 bit binary input B_0 , B_1 , B_2 , B_3 is applied to the data inputs D_0 , D_1 , D_2 , D_3 respectively of the four flip-flops. As soon as a negative clock edge is applied, the input binary bits will be loaded into the flip-flops simultaneously. The loaded bits will appear simultaneously to the output side. Only clock pulse is essential to load all the bits.

Block Diagram

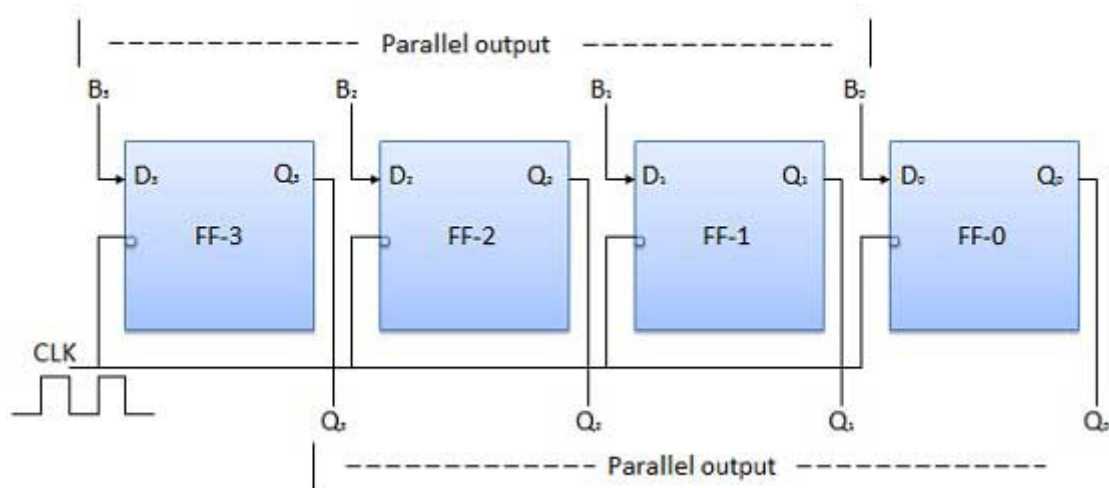


Fig.3.6.8 Block Diagram of Parallel Input Parallel Output (PIPO)

Parallel Input Serial Output (PISO)

Data bits are entered in parallel fashion.

The circuit shown below is a four bit parallel input serial output register.

Output of previous Flip Flop is connected to the input of the next one via a combinational circuit.

The binary input word B0, B1, B2, B3 is applied though the same combinational circuit.

There are two modes in which this circuit can work namely - shift mode or load mode.

Load mode

When the shift/load bar line is low (0), the AND gate 2, 4 and 6 become active they will pass B1, B2, B3 bits to the corresponding flip-flops. On the low going edge of clock, the binary input B0, B1, B2, B3 will get loaded into the corresponding flip-flops. Thus parallel loading takes place.

Shift mode

When the shift/load bar line is high (1), the AND gate 2, 4 and 6 become inactive. Hence the parallel loading of the data becomes impossible. But the AND gate 1,3 and 5 become active. Therefore the shifting of data from left to right bit by bit on application of clock pulses. Thus the parallel in serial out operation takes place.

Block Diagram

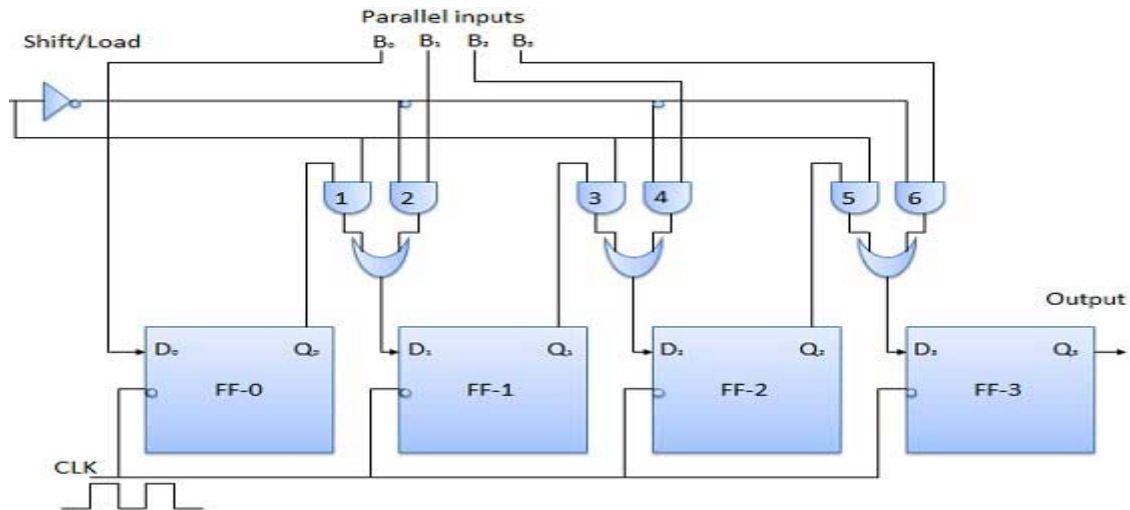


Fig.3.6.9 Block Diagram of Parallel Input Serial Output (PISO)

Let's check the take away from this lecture

1. The register is a type of _____

- a) **Sequential circuit**
- b) Combinational circuit
- c) CPU
- d) Latches

2. How many types of registers are?

- a) 2
- b) 3
- c) **4**
- d) 5

3. The main difference between a register and a counter is _____

- a) **A register has no specific sequence of states**
- b) A counter has no specific sequence of states
- c) A register has capability to store one bit of information but counter has n-bit

d) A register counts data

Exercise

Q.1 Explain about flow control security mechanism.

Q.2 Define covert channels.

Q.3 List the categories of covert channels and explain them.

Learning from this lecture: Learners will be able to understand the concept of Registers.

Conclusion

The study of flip flops explains the concept of different flip flops, their conversions, and use in digital logic design. Also by studying the excitation table, and timing diagrams helps to understand their use in design sequential circuits. The study of counters, generator and registers will be useful to design logical circuit for real time use.

Short Answer Questions:

1. What is Flip-flop?

Ans) Flip-flop is the common name given to two-state devices which offer basic memory for sequential logic operations. Flip-flops are heavily used for digital data storage and transfer and are commonly used in banks called "registers" for the storage of binary numerical data.

2. What is race around condition?

Ans) For J-K flip-flop, if $J=K=1$, and if $\text{clk}=1$ for a long period of time, then Q output will toggle as long as CLK is high, which makes the output of the flip-flop unstable or uncertain. This problem is called race around condition in J-K flip-flop.

3. What is Excitation Table?

Ans) An excitation table shows the minimum inputs that are necessary to generate a particular next state (in other words, to "excite" it to the next state) when the current state is known. They are similar to truth tables and state tables, but rearrange the data so that the

current state and next state are next to each other on the left-hand side of the table, and the inputs needed to make that state change happen are shown on the right side of the table.

4. What is SR flip flop?

Ans) SR flip-flop is basically a one-bit memory bistable device that has two inputs, one which will “SET” the device (meaning the output = “1”), and is labelled S and one which will “RESET” the device (meaning the output = “0”), labelled R.

5. What is JK flip flop?

Ans) The JK flip flop is basically a gated SR flip-flop with the addition of a clock input circuitry that prevents the illegal or invalid output condition that can occur when both inputs S and R are equal to logic level “1”.

Long Answer Questions:

1. Explain SR flip flop with excitation table and logic diagram.
2. Explain JK flip flop with excitation table and logic diagram.
3. Explain Master-Slave JK flip flop.
4. Convert SR flip flop to JK flip flop.
5. Convert JK flip flop to D flip flop.
6. Design of Asynchronous and Synchronous Counters.
7. Explain Modulus of the Counters.
8. Explain UP- DOWN counter.
9. Explain Ring and twisted ring/Johnson Counter.
10. Explain sequence generator
11. Write a short note on Registers: SISO, SIPO.
12. Write a short note on Registers: PIPO, PISO

Set of Questions for FA/IA/ESE

- Q. 1) State the difference between latch and flip flop.
- Q. 2) What is Excitation Table?
- Q. 3) State the difference between Truth table and Excitation table.
- Q. 4) What is race around condition?
- Q. 5) Convert D flip flop to T flip flop.

Module 3 Sequential Logic Design

- Q. 6) Explain Master-Slave JK flip flop.
- Q. 7) Convert SR flip flop to JK flip flop.
- Q. 8) Explain SR flip flop with excitation table and logic diagram.
- Q. 9) Explain JK flip flop with excitation table and logic diagram.
- Q. 10) Exercise UP- DOWN counter.
- Q.11) Explain Ring and twisted ring/Johnson Counter.
- Q.12) Explain sequence generator.
- Q. 13) Explain 4 Registers: SISO, SIPO, PIPO, PISO.

References:

1. Modern Digital Electronics, R. P. Jain McGraw Hill Fourth Edition 2010
2. Digital Logic and Computer Design, M. Morris Mano Pearson Sixth Edition 2017

Self-assessment

- Q.1) What is Flip flop? Define the role of flip flops as storage elements in digital logic design.
- Q. 2) Explain the Excitation tables for each flip flop.
- Q. 3) Describe the steps to convert from flip flop to another flip flop.
- Q. 4) Design of Asynchronous and Synchronous Counters.
- Q.5) Explain Modulus of the Counter.
- Q.6) Write a short note on counters: 1.UP- DOWN counter, 2.Ring and twisted ring/Johnson Counter.
- Q.7) Explain sequence generator.
- Q.8) Describe each Registers: SISO, SIPO, PIPO, PISO.

Self-evaluation

Name of Student	
Class	
Roll No.	

Module 3 Sequential Logic Design

Subject		
Module No.		
S.No		Tick Your choice
1.	Do you understand the concept of Flip Flop?	☐ Yes ☐ No
2.	Do you understand the concept of Concepts of Truth Tables,Excitation Tables and Race around condition	☐ Yes ☐ No
3.	Do you understand Master Slave J-K Flip Flops And how to create Timing Diagram	☐ Yes ☐ No
4.	Do you understand how to do Flip-flop conversion	☐ Yes ☐ No
5.	Do you understand the concept of Counters,sequence generator and Registers.	☐ Yes ☐ No
6.	Do you understand module 3?	☐ Yes, Completely. ☐ Partially. ☐ No, Not at all.